Amplitude Modulators

Circuits

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Basics of Modulators

Two approach of AM generation

1) Flywheel effect of Tank circuit (L and C) which has high Q factor.

2) Applying modulating signal and carrier signal to mixer (Non linear device).

Basics of Modulators

High Level Modulators: Modulation takes place in the final element of final stage where carrier is at maximum amplitude.

Low Level Modulators: Modulation takes place prior to the output element of the final stage of transmitter.

1) Nonlinear mixing approach

- Non-Linear Analog devi
- Frequency Translator

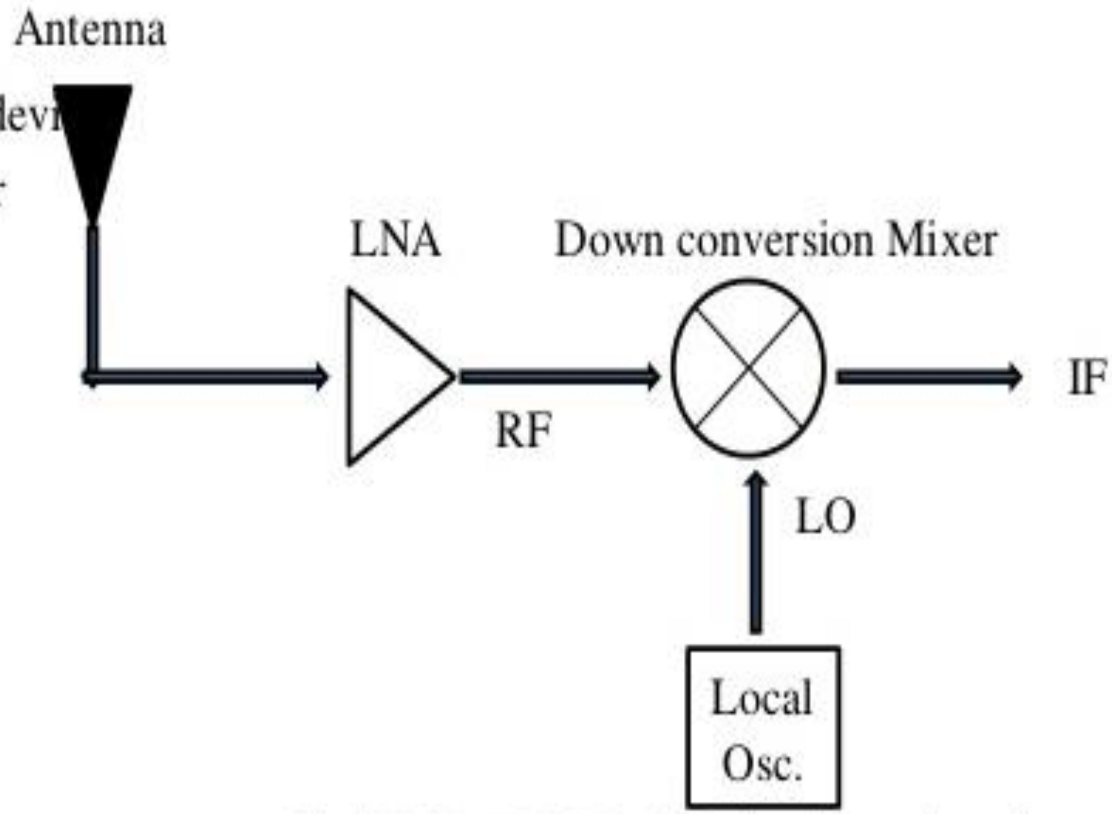
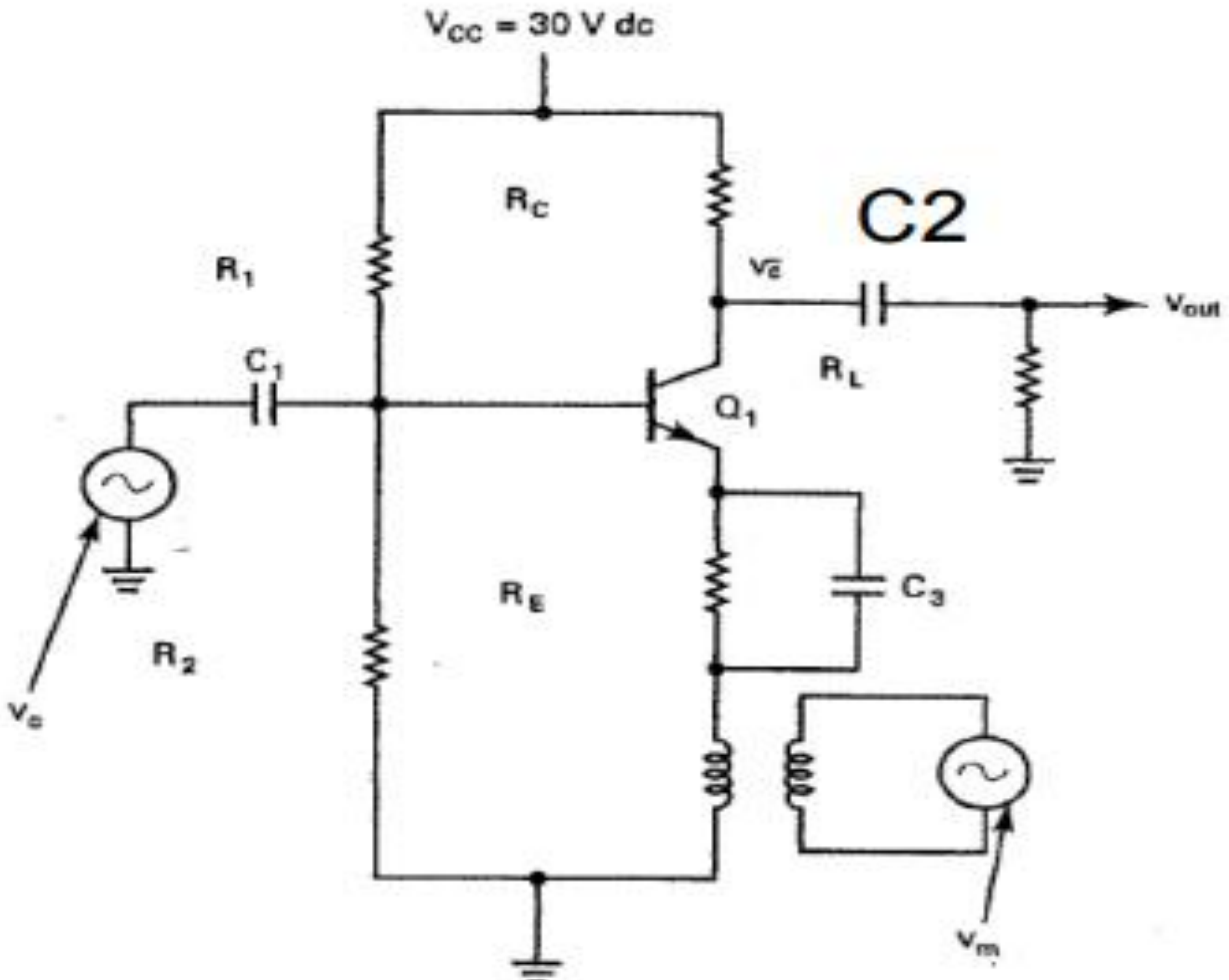


Fig.1: RF Front End / Role of mixer in generic receiver

If the inputs are sinusoids,
the ideal mixer output is the sum and difference frequencies given by

$$V_o = [A_1\cos(\omega_1 t)]\ [A_2\cos(\omega_2 t)] = \frac{A_1A_2}{2}[\cos(\omega_1 - \omega_2)t + [\cos(\omega_1 + \omega_2)t]$$

Low Level AM Modulator (Emitter)



Low Level Modulators (Emitter)

- If Vm(t) =0 → amplifier will be in **linear** mode
  - → $Aout=V_c cos(w_c t)$; Vc is voltage gain (unit less)
- If Vm(t) >0 → amplifier will be in **nonlinear** mode
  - → $Aout=[V_c + V_m cos(w_c t)] cos(w_c t)$
- Vm(t) is isolated using T1
  - The value of Vm(t) results in Q1 to go into cutoff or saturation modes
- C2 is used for coupling
  - Removes modulating frequency from AM waveform

Low Level Modulators

The Voltage gain is given by:

$$A_v = A_q\,[1 + m\sin(2\pi f_m t)]$$

where A_v = amplifier voltage gain with modulation (unitless)
A_q = amplifier quiescent (without modulation) voltage gain (unitless)

$\sin(2\pi f_m t)$ goes from a maximum value of $+1$ to a minimum value of -1.

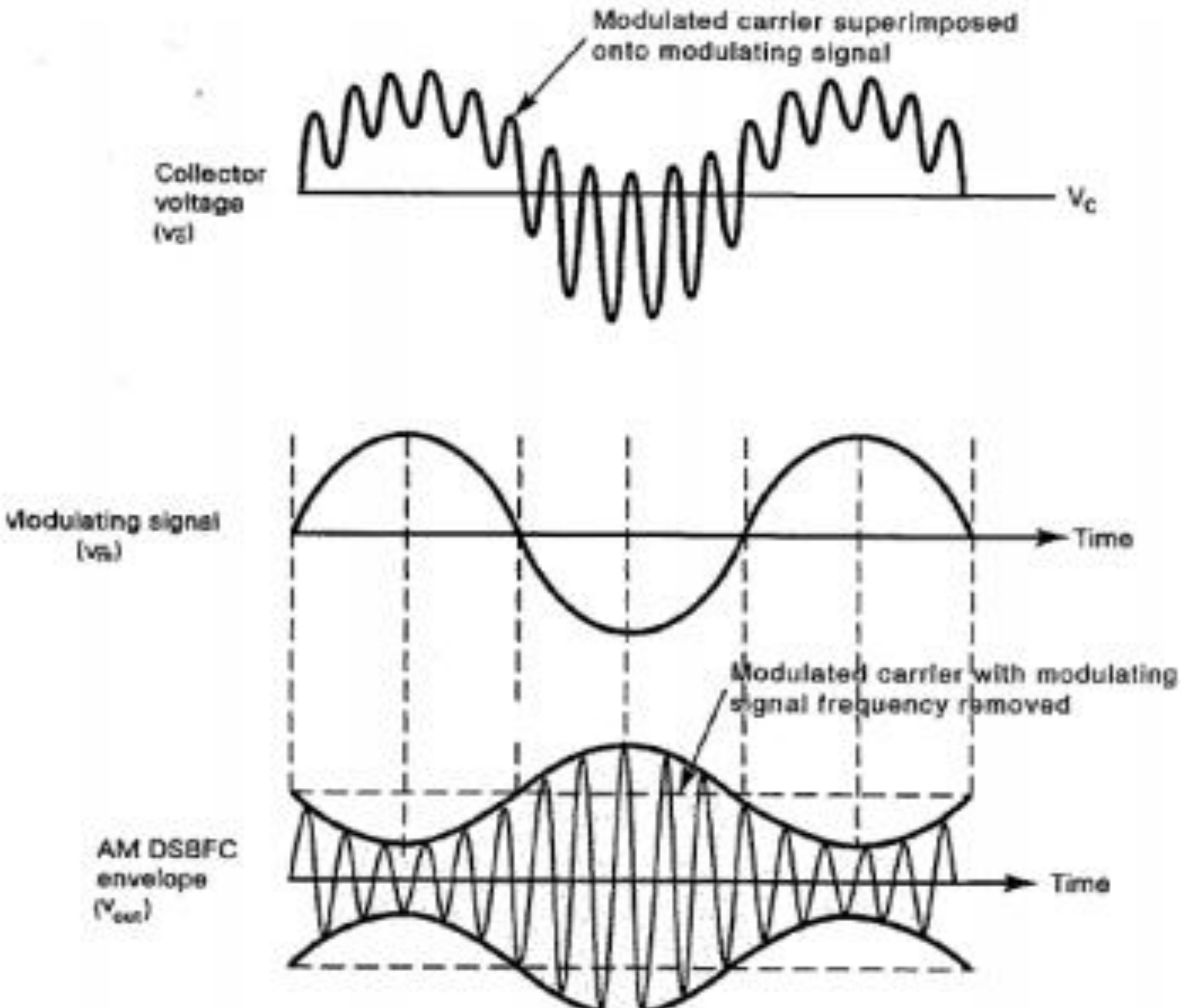
$$A_v = A_q(1 \pm m)$$

where m equals the modulation coefficient. At 100% modulation, $m = 1$,

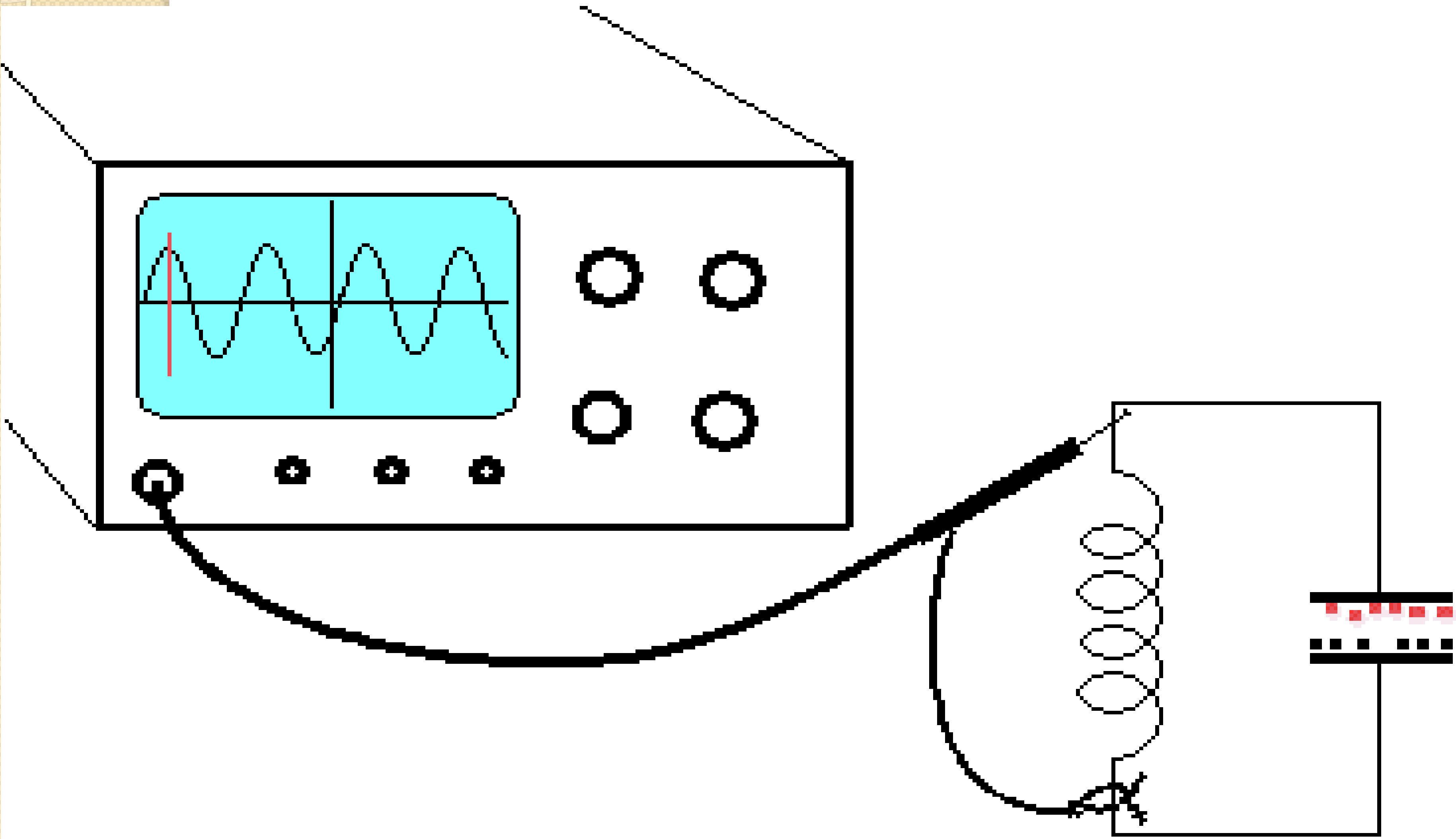
$$A_{v(max)} = 2A_q$$

$$A_{v(min)} = 0$$

Low Level AM Waveforms

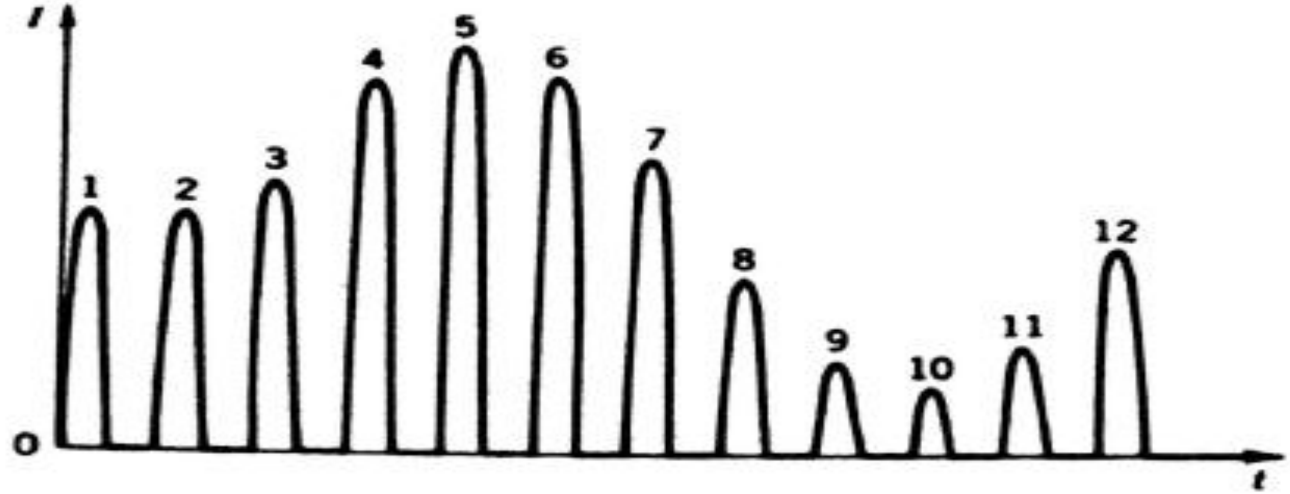


2) Flywheel effect of Tank circuit

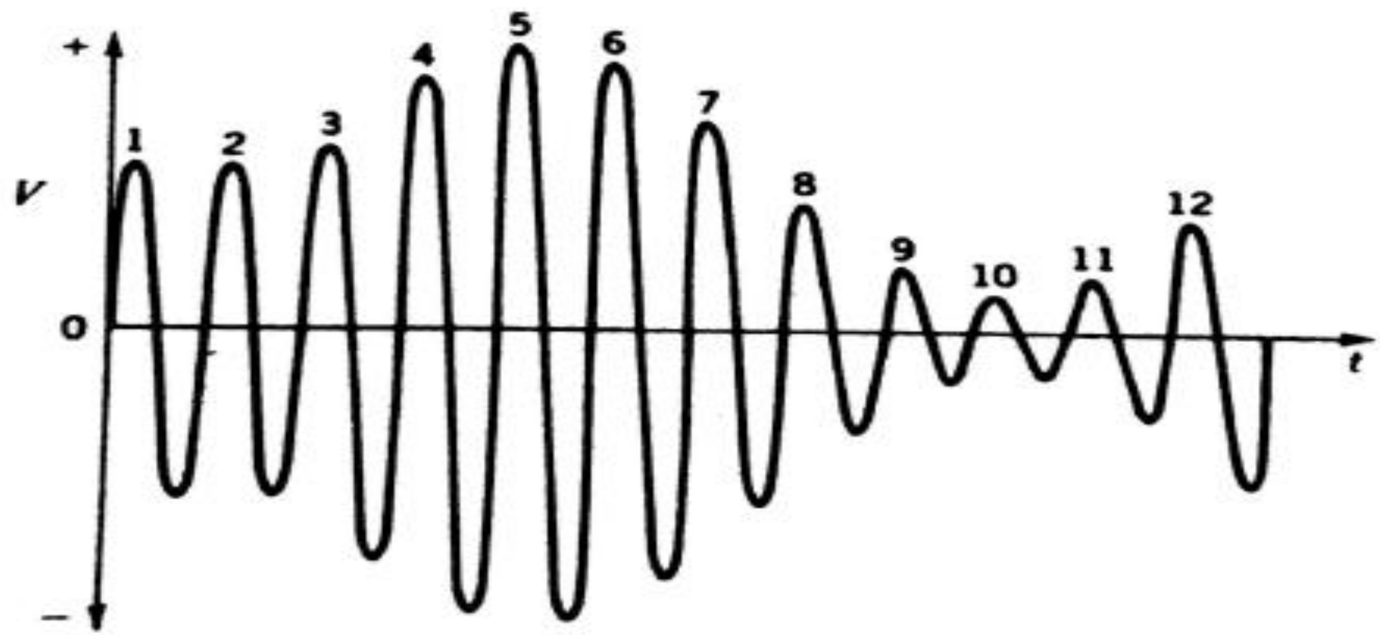


THE CRO MONITORING A TUNED CIRCUIT

Flywheel effect of Tank circuit

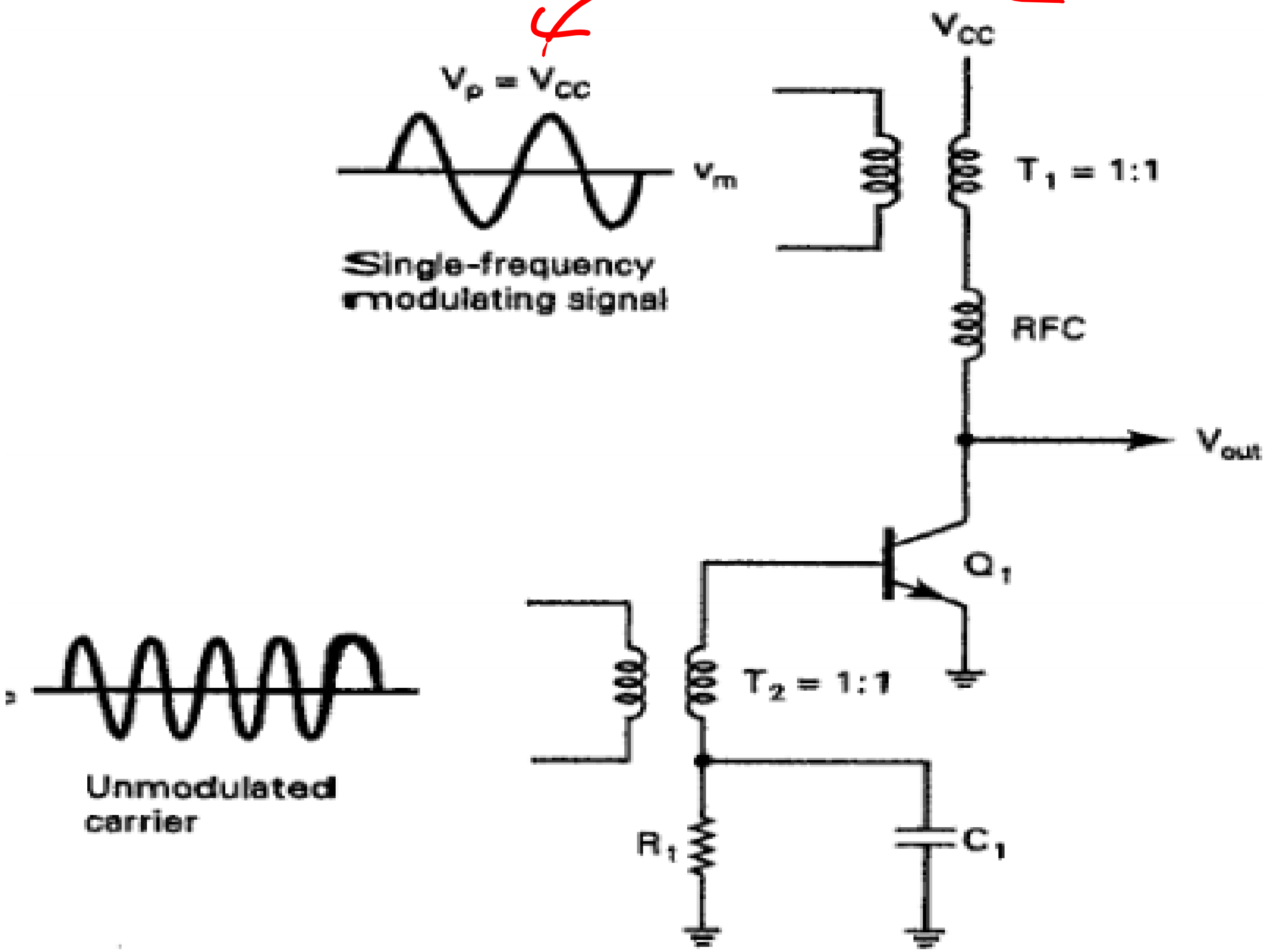


(a) Current pulses to tuned circuit



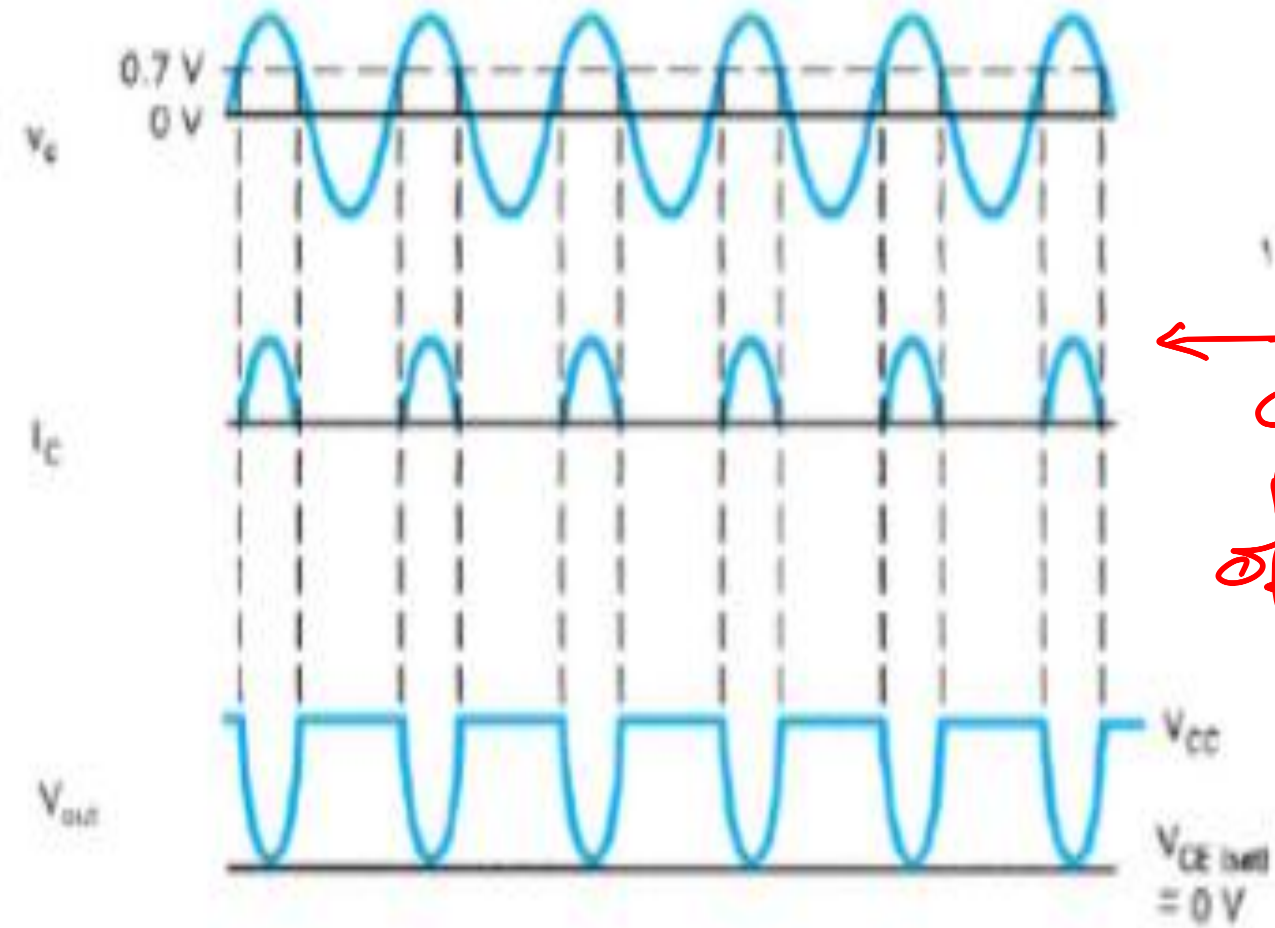
(b) Tuned-circuit AM voltage.

Medium Power AM Modulator



2Vcc

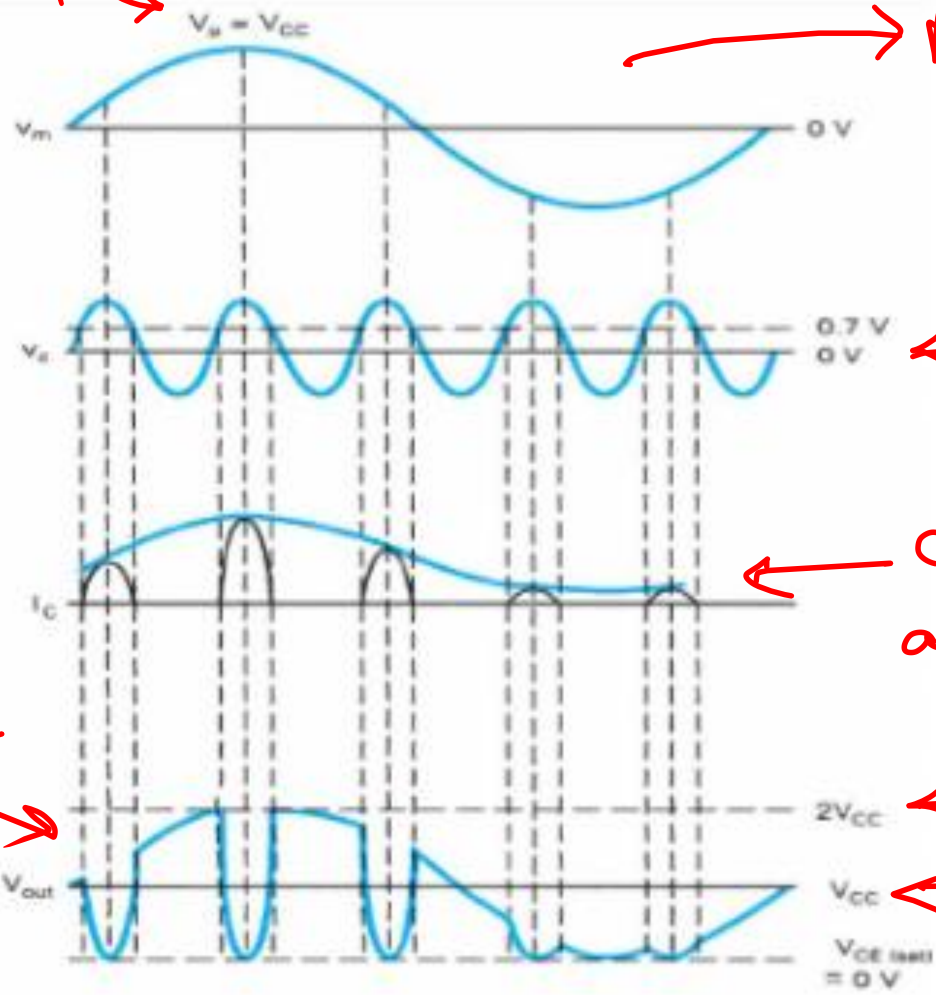
NO modulating signal



All current pulses are of equal

With Modulating Signal

$V_m = V_{cc}$



Modulating Signal

Voltage across collector

current pulse amplitude changes as per V_m

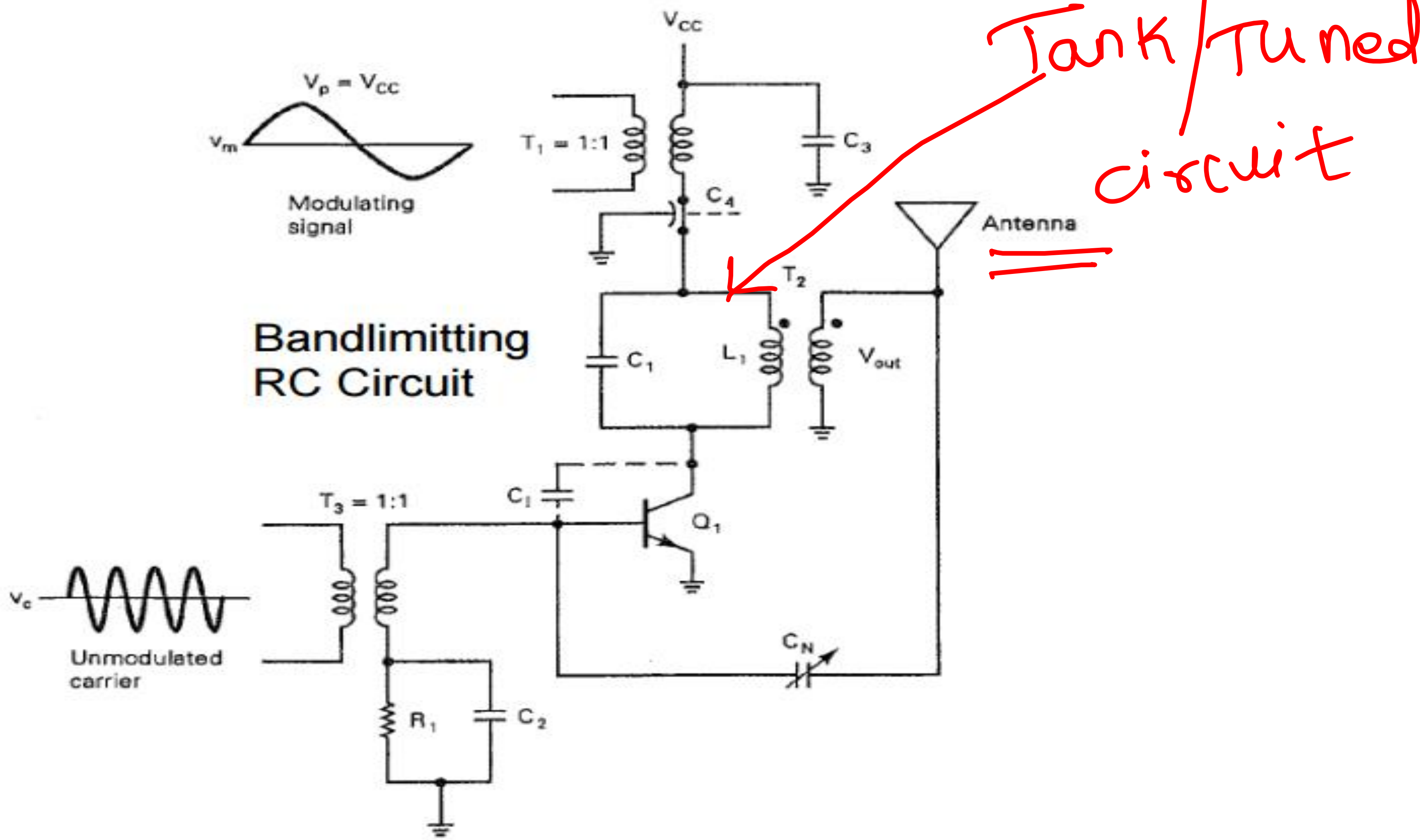
Max = 2VCC

Min = 0V

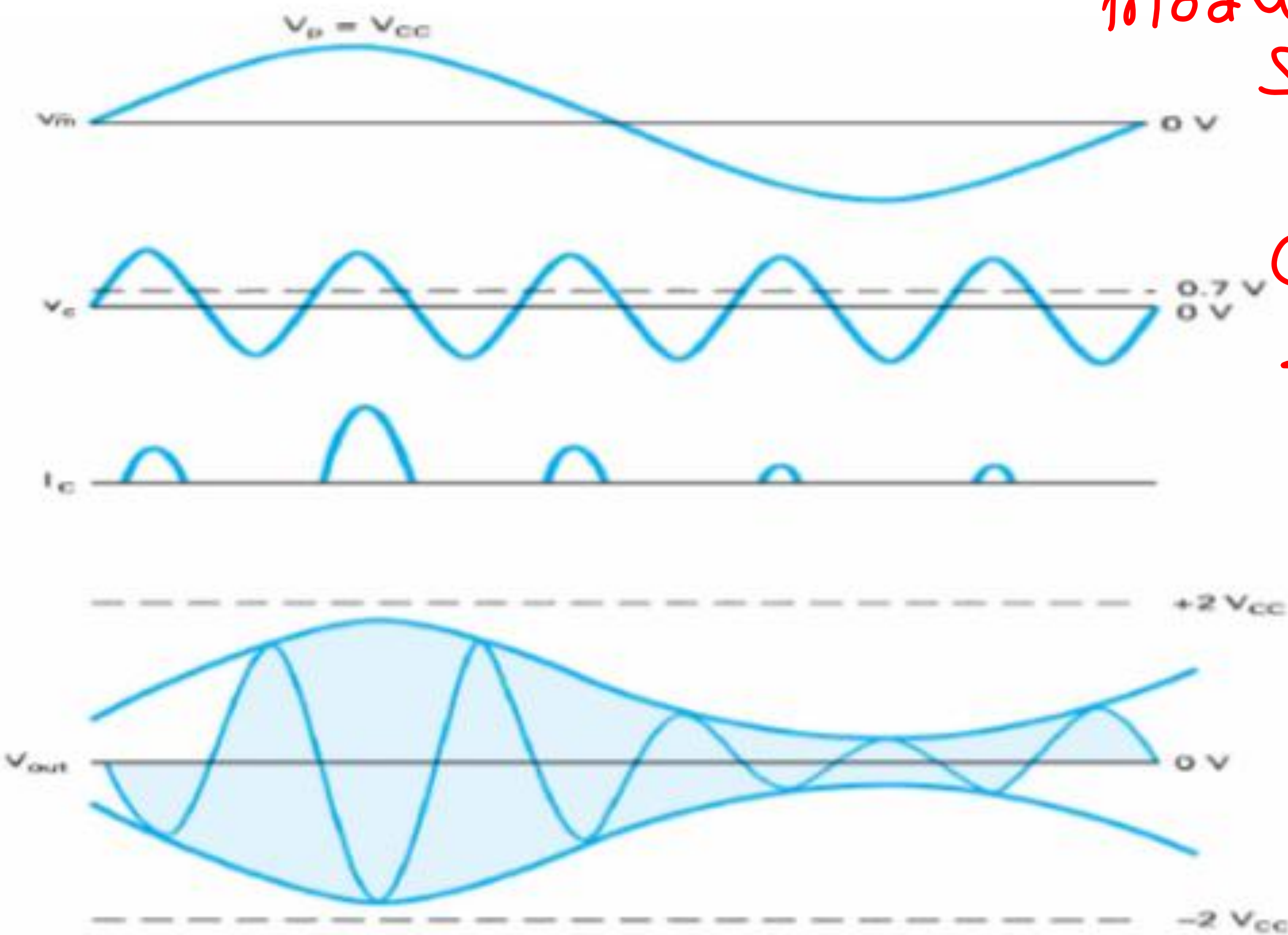
2VCC

0V

Medium Power AM Modulator(More Practical circuit)



Medium Power AM Modulator Waveforms



modulating signal

carrier signal

← Max

← min

Comparison of Emitter and Collector Modulator

Emitter modulator	Collector modulator
• Less symmetrical envelope	• More symmetrical envelope
• Less power efficiency	• Higher power efficiency
• Low output power	• Higher output power
• Require only small modulating signal drive power	• Need higher amplitude modulating signal
• Simpler circuit	• Complex circuit